



BAULKHAM HILLS HIGH SCHOOL

2017

YEAR 11 ASSESSMENT TASK 2

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 6-8
- Marks may be deducted for careless or badly arranged work

Total marks – 51

Exam consists of 5 pages.

This paper consists of TWO sections.

Section 1 – Page 2 (5 marks)

Questions 1 - 5

- Attempt Questions 1 - 5
Allow about 5 minutes for this section.

Section II – Pages 3 – 5 (46 marks)

- Attempt questions 6 - 8
Allow about 55 minutes for this section.

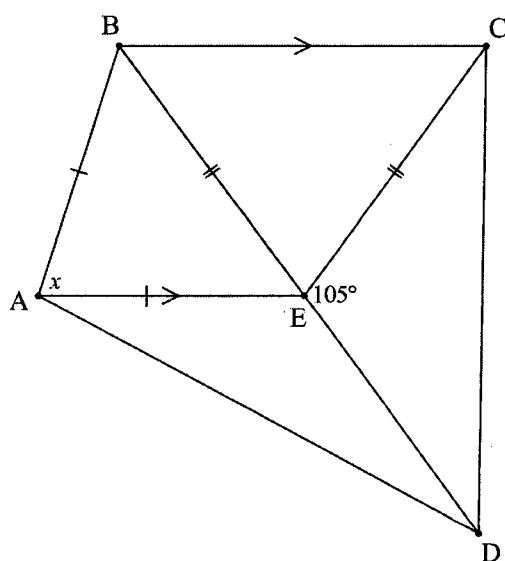
Section I - Multiple Choice (5 marks)

Allow about 5 minutes for this section.

Use the multiple choice page for Question 1 - 5

- 1 If $f(x) = 2x^3$, then $f'(2)$ equals
- (A) 8 (B) 16 (C) 24 (D) 144
- 2 A line passes through the origin and makes an angle of 45° with the positive direction of the x axis. The gradient of the line is
- (A) 0 (B) -1 (C) 1 (D) 45
- 3 Find $\lim_{x \rightarrow \infty} \frac{1 - x^2}{x^2}$
- (A) 1 (B) ∞ (C) 0 (D) -1
- 4 The line $y = mx + b$ is a tangent to the curve $y = x^3 - 3x + 2$ at the point $(-2, 0)$. What are the values of m and b ?
- (A) $m = 9$ and $b = -18$ (B) $m = 9$ and $b = 18$
(C) $m = 12$ and $b = -18$ (D) $m = 12$ and $b = 18$

- 5 The vertices of quadrilateral $ABCD$ are joined at E such that $BC \parallel AE$, BE is produced to D . $\angle CED = 105^\circ$, $BE = CE$ and $AB = AE$. Determine the size of x .



Not to Scale

- (A) 105°
(B) 85°
(C) 75°
(D) 52.5°

End of Section 1

Section II (46 marks)**Allow about 55 minutes for this section.****Answer each question on the appropriate page in the writing booklet.****Question 6 (15 marks)****Marks**

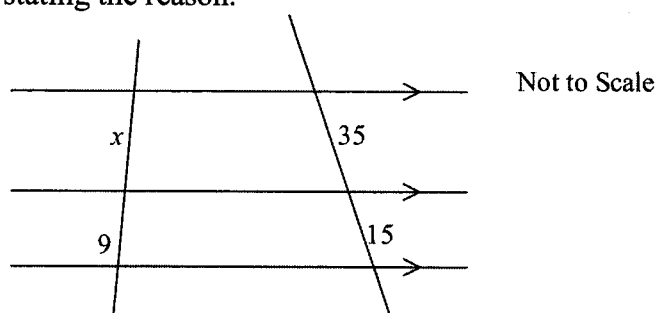
- a) Find the gradient of the normal to the curve $y = 3 - x^2$ at $x = -1$

3

- b) Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x - 3}$

2

- c) Find the value of x , stating the reason.

2

- d) Differentiate with respect to x :

(i) $y = \frac{6x^3 - 4x^2}{2x}$

2

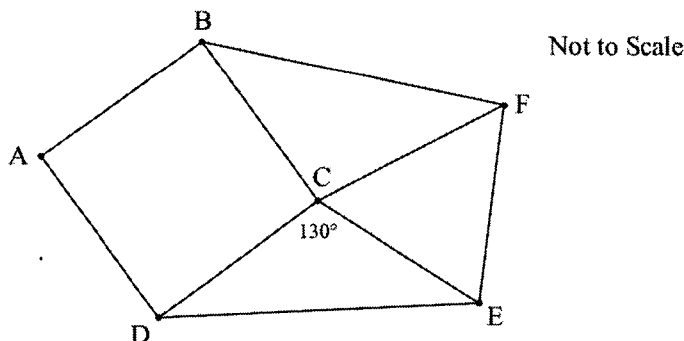
(ii) $f(x) = (3x - 5)(2x + 5)$

2

- e) Find the equation of the straight line passing through the point (1,5) and through the point of intersection of the lines $3x - 4y + 2 = 0$ and $5x + 2y = 14$.
Give your answer in simplest general form.

3

- f) In the figure below, $ABCD$ is a square, CEF is an equilateral triangle, $\angle DCE = 130^\circ$ and $DC = CE$.

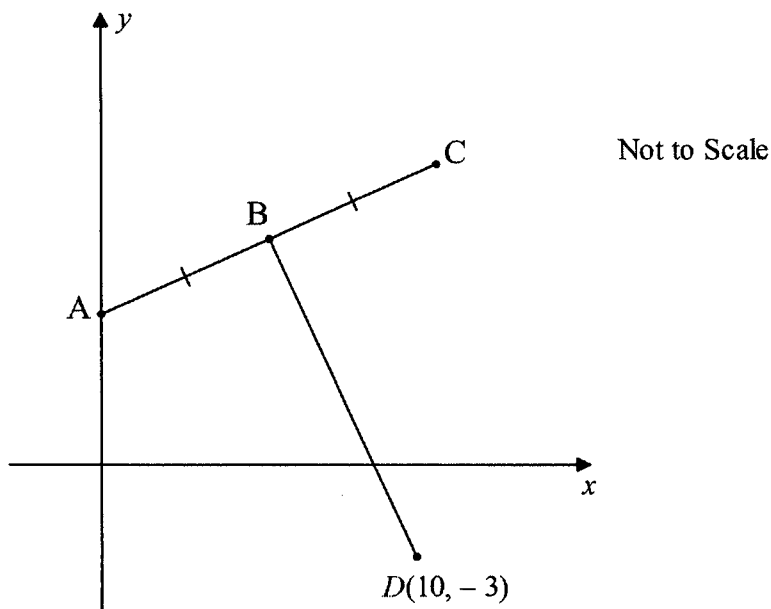
1

Find the size of $\angle CBF$ (without giving reasons)

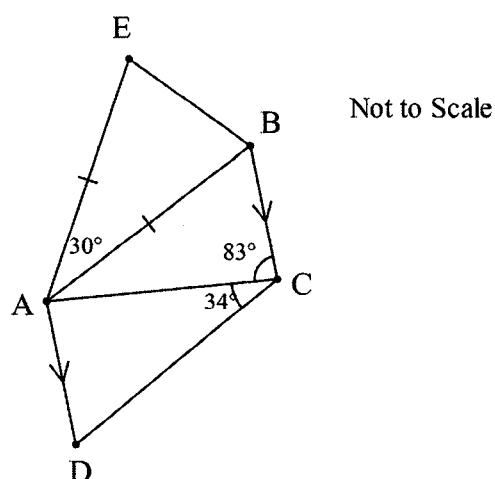
End of Question 6

Question 7 (15 marks)

- a) Find the derivative of $f(x) = 5x - 2x^2$ by first principles. 3
- b) The diagram shows points A, B and C lying on the line $2y = x + 4$. The point A lies on the y -axis and $AB = BC$. The line from $D(10, -3)$ to B is perpendicular to AC .



- (i) Find the coordinates of A . 1
- (ii) Find the equation of the line BD . 2
- (iii) Find the coordinates of C . 3
- c) In the diagram below: $AD \parallel BC$, $AE = AB$, $\angle BAE = 30^\circ$, $\angle BCA = 83^\circ$, $\angle ACD = 34^\circ$ and $\angle EBC = 138^\circ$.

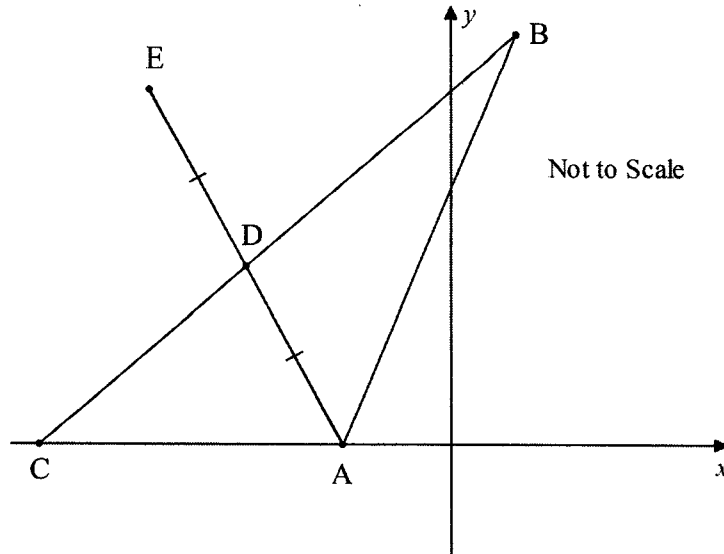


- (i) Prove that $AB \parallel DC$ 3
- (ii) Prove that $\triangle ABC \equiv \triangle ACD$ 3

End of Question 7

Question 8 (16 marks)

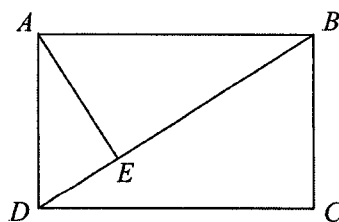
- a) In the diagram A, B and C are $(-1,0)$, $(2,4)$ and $(-6,0)$ respectively. D is the midpoint of AE and has coordinates $(-2,2)$.



Copy the diagram into your booklet and include the information above.

- | | | |
|-------|---|---|
| (i) | Find the length of the interval AD . | 1 |
| (ii) | Show that D is the midpoint of BC . | 1 |
| (iii) | Show that the equation of the line BC is $x - 2y + 6 = 0$ | 2 |
| (iv) | Find the perpendicular distance of A from the line BC . | 2 |
| (v) | What type of quadrilateral is $ABEC$? (Give reasons for your answer) | 2 |
| (vi) | Find the area of this quadrilateral. | 2 |

- b) $ABCD$ is a rectangle and $AE \perp BD$.
 $AE = 5\text{cm}$ and $DE = 2\text{ cm}$.



- | | | |
|-------|--|---|
| (i) | Copy the diagram into your booklet and prove that triangles AED and BCD are similar. | 2 |
| (ii) | Hence, show that $AD^2 = BD \times DE$. | 2 |
| (iii) | Find the area of $ABCD$. | 2 |

End of Exam

multi choice.

1. C 2. C 3. D 4. B 5. C

1. $f(x) = 2x^3$

$f'(x) = 6x^2$ $f'(2) = 6 \times 4 = 24$ (C)

2. $\tan \theta = m$

$\tan 45 = 1$ (C)

3. $\lim_{x \rightarrow \infty} \frac{1-x^2}{x^2} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x^2} - \frac{x^2}{x^2}}{\frac{x^2}{x^2}}$

$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x^2} - 1}{1}$

as $x \rightarrow \infty$ $\frac{1}{x^2} \rightarrow 0$ (D)

$= -1$

4. $y = x^3 - 3x + 2$

$\frac{dy}{dx} = 3x^2 - 3$

at $x = -2$ $y = 9$ $m = 9$ $\therefore y = 9x + b$ $0 = -18 + b$ $b = 18$ $\therefore y = 9x + 18$ (B)

5. $\triangle$ sum isosceles triangle with BED a straight L. (C)

Question 6.

a) $y = 3 - x^2$

$\frac{dy}{dx} = -2x$

at $x = -1$ $m = 2$ $\therefore$ normal $m_2 = -\frac{1}{m_1} = -\frac{1}{2}$

b) $\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x - 3} = \lim_{x \rightarrow 3} \frac{(x-3)(x+1)}{x-3}$

$= 4$

c) $\frac{x}{9} = \frac{35}{15}$ ratio of intercepts on parallel lines

$x = 21$ units

quest 6 cont.

d) $y = \frac{6x^3 - 4x^2}{2x}$ quotient rule
 $= 3x^2 - 2x$ or
 $\frac{dy}{dx} = 6x - 2$

ii) $f(x) = (3x - 5)(2x + 5)$

$= 6x^2 + 5x - 25$

$f'(x) = 12x + 5$

e) Either method.

K method.

$3x - 4y + 2 + k(5x + 2y - 14) = 0$

passing through (1,5)

$3 - 20 + 2 + k(5 + 10 - 14) = 0$

$-15 + k(15 - 14) = 0$

$k = 15$

$\therefore 3x - 4y + 2 + 15(5x + 2y - 14) = 0$

$3x - 4y + 2 + 75x + 30y - 210 = 0$

$78x + 26y - 208 = 0$

$3x + y - 8 = 0$

or

pt of intersection

$3x - 4y = -2$ (1)

$5x + 2y = 14$ (2)

$10x + 4y = 28$ (3)

(5) + (1) $13x = 26$

$x = 2$

sub in (1)

$6 - 4y = -2$

$-4y = -8$

$y = 2$

$\therefore (2, 2)$

$\therefore$ gradient for (2,2) and (1,5)

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{5 - 2}{1 - 2} = -3$

$\therefore$ new line $y - y_1 = m(x - x_1)$

$y - 2 = -3(x - 2)$

$3x + y - 8 = 0$

2 marks correct answer.
1 mark 1 mistake.

2 marks correct answer
1 mark 1 mistake

3 marks correct method with correct equation
2 marks one error in process and simplified general form or correct process not simplified general form.
1 mark working towards solution 2 mistakes

1 mark $\angle CBF = 50^\circ$

Question 7.

a) $f(x) = 5x - 2x^2$
 $f(x+h) = 5(x+h) - 2(x+h)^2$
 $= 5x + 5h - 2x^2 - 4xh - 2h^2$
 $f(x+h) - f(x) = 5h - 4xh - 2h^2$
 $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
 $= \lim_{h \rightarrow 0} \frac{h(5 - 4x - 2h)}{h}$
 $= 5 - 4x$

b) i) AC: $2y = x + 4$
 at A: $x = 0$ $\therefore 2y = 4$ A is (0, 2)
 $y = 2$

ii) AC: $y = \frac{1}{2}x + 2$ $m_1 = \frac{1}{2}$
 BD: $m_2 = -\frac{1}{m_1}$
 $= -2$ point D (10, -3)
 $y - y_1 = m(x - x_1)$
 $y + 3 = -2(x - 10)$
 $y = -2x + 17$
 $2x + y - 17 = 0$

iii) find B
 BD: $y = -2x + 17$ --- ①
 AC: $y = \frac{1}{2}x + 2$ --- ②
 sub ② into ①
 $\frac{1}{2}x + 2 = -2x + 17$
 $x + 4 = -4x + 34$
 $5x = 30$
 $x = 6$
 sub into ② $y = \frac{1}{2} \times 6 + 2$
 $= 5$
 B (6, 5)

quest 7 Cont.

B is the midpt of AC
 $\therefore x = \frac{x_1 + x_2}{2}$ $y = \frac{y_1 + y_2}{2}$
 $b = \frac{0 + x_2}{2}$ $5 = \frac{2 + y_2}{2}$
 $x_2 = 12$ $y_2 = 8$ C is (12, 8)

c) i) Aim: to prove $AB \parallel DC$.
 Method: In $\triangle AEB$
 $\angle AEB = \angle ABE$ (Ls opposite equal sides are equal)
 $2x + 30^\circ = 180^\circ$ (L sum $\triangle$)
 $x = 75^\circ$
 now $\angle EBC = 138^\circ$ (given)
 $\therefore \angle ABC = 138^\circ - \angle ABE$ (adjacent Ls)
 $= 63^\circ$
 $\therefore \angle ABC + \angle BCD = 63^\circ + 83^\circ + 34^\circ$
 $= 180^\circ$
 $\therefore AB \parallel DC$ (Co interior Ls supplementary)

ii) Aim: prove $\triangle ABC \equiv \triangle ACD$
 Method: In $\triangle ABC$ and $\triangle ACD$
 $\angle BAC = \angle DCA$ (alternate Ls, $AB \parallel DC$)
 $= 34^\circ$
 $\angle BCA = \angle DAC$ (alternate Ls, $AD \parallel BC$)
 $= 83^\circ$
 AC is common
 $\angle ADC = 63^\circ$ (L sum $\triangle ACD = 180^\circ$)
 $\therefore \angle ADC = \angle ABC$ (from i)
 $\therefore \triangle ABC \equiv \triangle ACD$ (S, A, A)

3 marks
 clear setting out
 showing the
 correct answer.

2 marks
 error with placement
 of lim or another
 mistake

1 mark...
 2 mistakes

1 mark
 correct
 response.

2 marks
 correct formula
 correct answer

1 mark
 either correct
 gradient or
 correct equation
 from incorrect
 gradient

3 marks
 correct point
 from clear
 working out

2 marks
 correct value
 for B or
 correct method
 with one mistake

1 mark
 2 mistakes with
 appropriate
 method.

3 marks
 correctly
 deduces answer.

2 marks
 one error in
 reasoning

1 mark
 two answers
 in reasoning

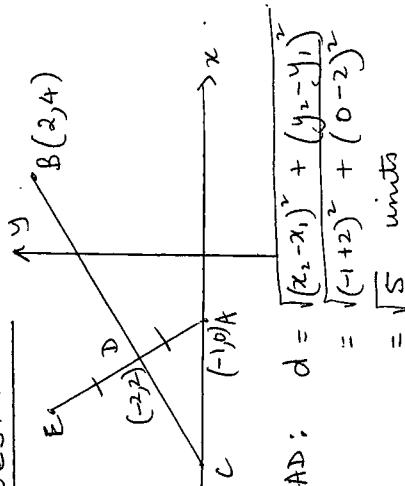
3 marks
 well set out
 proof.

2 marks
 one error
 in reasoning

1 mark
 two errors
 in reasoning

Question 8.

a)



$$\begin{aligned} \text{ii) AD: } d &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(-1 - 2)^2 + (0 - 2)^2} \\ &= \sqrt{5} \text{ units} \end{aligned}$$

iii) D

$$\begin{aligned} \text{Mid pt BC: } x &= \frac{x_1 + x_2}{2} = \frac{-1 + 2}{2} = \frac{1}{2} \\ y &= \frac{y_1 + y_2}{2} = \frac{0 + 4}{2} = 2 \end{aligned}$$

$\therefore (-2, 2)$ which is D.

iv) BC: $m = \frac{y_2 - y_1}{x_2 - x_1}$

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 4 &= \frac{1}{2}(x - 2) \\ 2y - 8 &= x - 2 \\ x - 2y + 6 &= 0 \end{aligned}$$

use pt (2, 4)

$$\text{v) A}(-1, 0), \text{ perp d} = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$\begin{aligned} &= \frac{|-1 \times 1 - 2 \times 0 + 6|}{\sqrt{1 + 4}} \\ &= \frac{5}{\sqrt{5}} \text{ or } \sqrt{5} \text{ units} \end{aligned}$$

v) ABEC is a rhombus - diagonals perpendicular bisect through D.

[- find lengths AB and CA - adjacent sides equal - diagonals bisect at D.]

vi) Area = $\frac{1}{2} \times \text{diagonals}$ or $2 \times \triangle ACB$.

$$\triangle ACB = \frac{1}{2} \times 5 \times 4 = 10$$

$$\text{Area ABEC} = 2 \times 10 = 20 \text{ units}^2$$

1 mark was deducted for no diagram

1 Mark correct response

1 Mark correct response but connection should be clear linking midpt to D

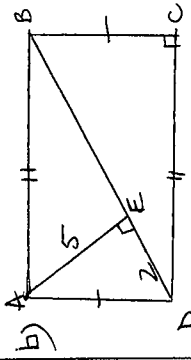
2 Marks 1 mark m value 1 mark equation in general form.

2 Marks 2 marks clear substitution into formula and correct answer 1 mark calculator error

2 marks 1 for Rhombus 1 for supporting reasons.

2 marks clear method from part (v).

Quest 8 Cont.



i) Aim: prove $\triangle AED \cong \triangle BEC$

Method: In $\triangle AED$ and $\triangle BEC$

$AE \perp BD$ (given)

$\therefore \angle AED = 90^\circ$

$\angle BCD = 90^\circ$ (L in a rectangle)

$= \angle AED$

$\angle ADE = \angle CBE$ (alternate $\angle$ s, $AD \parallel BC$)

$\therefore \triangle AED \cong \triangle BEC$ (matching is equal or A.A.)

ii) $\frac{AD}{BD} = \frac{DE}{BC}$ (matching sides in ratio)

but $AD = BC$ (equal sides in a rectangle)

$\therefore AD \times AD = BD \times DE$

$AD^2 = BD \times DE$

$$\begin{aligned} \text{iii) } AD &= \sqrt{AE^2 + DE^2} \quad (\text{Pythagoras' theorem}) \\ &= \sqrt{25 + 4} \\ &= \sqrt{29} \end{aligned}$$

from (ii)

$$AD^2 = BD \times DE$$

$$= BD \times 2$$

$$\therefore \frac{29}{2} = BD$$

$$\text{Area } ABCD = 2 \times \text{Area } \triangle ADB$$

$$= 2 \times \frac{1}{2} \times 5 \times \frac{29}{2}$$

$$= 72.5 \text{ cm}^2$$

2 Marks

2 clear steps with reasons

1 Mark

error with reasoning

2 Marks

2 reasons

with correct ratio

1 mark

working not clearly justified

2 Marks

correct answer

1 Mark

BD correct or equivalent main value.

